

FAIRFORD, United Kingdom

WMO: 036440

Lat: 51.682N

Lon: 1.790W

Elev: 87

StdP: 100.28

Time Zone: 0.00 (EUW)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-3.0	-6.1	2.3	-2.5	-4.8	2.6	-1.2	14.1	9.6	12.8	9.8	1.7	250	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	26.5	18.2	24.2	17.5	22.7	16.7	19.8	23.8	18.6	22.2	17.7	20.9	4.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	20.6	17.4	12.6	19.4	16.6	11.9	18.7	56.8	24.1	53.1	22.5	50.2	21.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.3	9.0	DB	-7.0	29.8	3.4	2.2	-9.4	31.4	-11.4	32.7	-13.3	33.9	-15.7	35.6	(4)
(5)			WB	-6.6	21.4	2.4	1.4	-8.4	22.5	-9.8	23.3	-11.1	24.1	-12.9	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	4.2	4.9	6.7	9.0	12.3	15.1	17.4	16.4	14.3	11.4	7.2	4.7
	DBStd	5.44	3.44	3.19	2.79	2.93	2.73	2.50	2.63	2.11	2.49	2.70	3.25	3.88
	HDD10.0	769	181	145	106	52	8	0	0	0	1	16	93	168
	HDD18.3	2959	438	378	360	279	189	102	51	68	123	215	333	423
	CDD10.0	887	1	1	4	22	78	152	228	197	130	59	9	4
	CDD18.3	34	0	0	0	0	1	4	21	7	2	0	0	0
	CDH23.3	298	0	0	0	1	9	51	183	44	9	0	0	0
	CDH26.7	54	0	0	0	0	0	11	39	4	0	0	0	0
(14) Wind	WSAvg	4.1	4.5	4.4	4.5	4.0	4.3	3.9	3.9	3.9	3.6	3.8	3.9	4.3
(15) Precipitation	PrecAvg	777	82	57	50	55	61	55	57	63	57	77	83	79
	PrecMax	1092	184	130	96	161	132	154	170	126	111	142	146	139
	PrecMin	552	10	8	13	3	7	11	7	3	15	27	30	20
	PrecStd	137	38	32	23	35	28	33	34	36	30	33	33	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.8	17.8	22.1	24.9	28.2	30.4	27.1	24.8	19.2	15.9	14.1
		MCWB	11.5	9.9	11.2	13.9	17.5	19.1	20.9	18.9	17.9	16.5	14.7	13.6
	2%	DB	11.8	11.6	14.8	19.1	22.0	24.3	27.2	24.2	22.2	17.8	14.2	13.0
		MCWB	11.2	10.2	11.3	13.3	15.8	17.4	18.3	17.8	17.4	15.5	13.3	12.8
	5%	DB	10.8	10.8	13.0	16.8	19.9	22.1	25.1	22.5	20.6	16.8	13.2	12.0
		MCWB	10.1	9.7	9.8	11.9	14.5	16.4	17.4	17.0	16.1	15.0	12.3	11.4
	10%	DB	9.8	9.9	11.5	14.8	17.8	20.2	23.0	21.0	19.1	15.8	12.3	10.8
		MCWB	9.3	9.0	9.1	11.2	13.4	15.5	16.6	16.4	15.5	14.2	11.4	10.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.8	13.0	15.5	18.3	20.5	21.6	20.4	20.9	17.5	15.2	13.7
		MCDB	12.3	12.3	14.1	20.0	22.3	26.4	27.9	23.9	22.4	18.7	15.4	13.8
	2%	WB	11.1	10.7	11.7	14.0	16.6	18.7	19.8	19.0	18.5	16.3	13.8	12.8
		MCDB	11.3	11.2	13.5	17.7	20.6	22.3	24.5	22.1	20.4	17.2	14.1	13.0
	5%	WB	10.1	10.0	10.5	12.9	15.2	17.2	18.5	18.1	17.3	15.5	12.8	11.6
		MCDB	10.4	10.5	12.2	15.6	18.4	20.7	23.3	21.0	19.3	16.3	13.3	11.8
	10%	WB	9.1	9.0	9.5	11.8	14.3	16.2	17.5	17.3	16.2	14.6	11.7	10.3
		MCDB	9.4	9.6	11.0	14.3	16.9	19.4	21.7	19.9	18.4	15.5	12.4	10.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.5	8.0	9.6	9.3	9.0	9.8	8.8	8.9	6.9	6.5	6.2
		MCDBR	6.2	7.3	9.9	13.2	12.5	12.1	13.0	11.0	10.4	7.3	6.6	5.8
		MCWBR	6.3	5.7	6.8	7.7	6.9	6.3	5.7	5.5	6.1	5.2	6.0	5.4
	5% WB	MCDBR	6.2	6.6	9.1	11.4	10.3	10.4	11.2	9.1	8.6	6.3	6.5	5.7
		MCWBR	6.3	5.7	7.1	7.2	6.4	6.1	5.8	5.5	6.3	5.1	6.3	5.8
(40) Clear-Sky Solar Irradiance	taub		0.324	0.339	0.374	0.396	0.401	0.396	0.393	0.382	0.367	0.352	0.325	0.313
	taud		2.354	2.370	2.271	2.224	2.246	2.288	2.306	2.359	2.396	2.418	2.403	2.333
	Ebn at Noon		677	777	813	838	852	858	854	845	816	752	672	619
	Edh at Noon		63	81	107	126	129	125	121	109	94	76	60	55
(44) All-Sky Solar Radiation	RadAvg		0.80	1.46	2.55	3.93	4.80	5.25	4.96	4.12	3.10	1.79	0.95	0.63
	RadStd		0.07	0.18	0.34	0.46	0.29	0.54	0.50	0.34	0.26	0.16	0.10	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page